See what matches. Prove what you checked.

Pre-publication creative-rights evidence. Three independent AI lanes answer three different questions about an image, your policy turns them into one decision, and the whole result is sealed into a receipt that anyone can verify without trusting us.

LANE 1 · COPY

Does this reuse a work someone already registered?

LANE 2 · ORIGIN

Was a generative model involved in making it?

LANE 3 · PROFILE

Does it resemble a specific creator's body of work?

Project
CreatorProof v0.10.0
Build MODEL-ACCURACY-HARDENING-2026.08.10

Document
Problem · Solution · Users · Technology · AI
Prepared August 2026

00 At a glance

Project name	**CreatorProof** — visual rights intelligence
Team name	[TEAM NAME]
Members	[MEMBER 1 — role] · [MEMBER 2 — role] · [MEMBER 3 — role]
One line	Pre-publication creative-rights evidence for agencies, brands and marketplaces — plus a receipt you can verify yourself.
Category	Applied AI · Blockchain-anchored provenance · Creative-industry infrastructure
Stage	Working end-to-end prototype. Not a mockup, not a slide deck.
Demo	[DEMO URL] · Artist portal `/artist` · Checker portal `/user`
Source	[GITHUB URL]
Video pitch	[VIDEO URL]

What actually runs, in numbers

3
independent AI evidence lanes that are never collapsed into one score

3
neural models loaded and running locally on device

400
automated tests across 47 suites, all passing

~46k
lines of application, test and interface code

98.1%
match recall on a 364-case internal copy-detection benchmark

0
false matches across 115 unrelated-image negative pairs

32
bytes written on chain per attestation — a hash, never the artwork

0
legal judgments claimed, by design

> **A NOTE ON THE NUMBERS ABOVE**
>
> The benchmark figures come from a harness we wrote and ran ourselves (`scripts/copy_detection_bench.py`), against 26 real-world transformations applied to 14 source images. They are honest measurements of our own pipeline, not results on a public leaderboard dataset, and we describe them that way throughout this document. Section 06 gives the full methodology and the cases we still fail.

01 Problem statement

Every organisation that publishes images now has to answer three questions before it presses publish — and the entire market sells them one answer.

A brand team receives a campaign image from an agency. A marketplace receives an upload from a seller. A publisher receives a submission from a freelancer. Before that image goes out under their name, somebody has to decide whether it is safe. Today that decision is made by a person squinting at a picture, running a reverse image search, and hoping.

The uncomfortable truth is that "is this safe to publish?" is not one question. It is three:

REUSE

Does this image reuse a work that someone has already registered?

Not "does it look similar" — does the actual content of a known work appear inside this file, possibly cropped, mirrored, re-compressed, screenshotted or pasted into a social post?

ORIGIN

Was a generative model involved in producing it?

A question with real contractual and regulatory weight — many client contracts and platform policies now require disclosure — and one that no detector on earth answers with certainty.

RESEMBLANCE

Does it resemble a specific living creator's body of work?

The style-mimicry question that generative models made urgent, and the one most likely to end in a public dispute even when no copying occurred.

1.1 Why the single-number approach fails

The tools on the market answer one of these and imply the others. An "AI detector" returns *87% AI* and a user reads it as a verdict. A reverse-image tool returns a similarity percentage and a user reads it as infringement. A style-matching tool returns a creator name and a user reads it as authorship. Each collapse is a different kind of harm:

- **Collapsing origin into guilt** punishes legitimate AI-assisted workflows and produces false accusations against human artists whose work happens to score high.
- **Collapsing similarity into infringement** produces legal claims that do not survive contact with a lawyer, and burns the relationship with the creator.
- **Collapsing style into authorship** is the most dangerous of the three: resemblance is not evidence of copying, and treating it as such defames the accused.

There is also a fourth failure that nobody talks about: **silence read as safety.** When a tool searches an incomplete database and returns "no matches found", the user hears "this image is clean". It is not. It means the tool did not look everywhere. That single ambiguity is responsible for a large share of the false confidence in this market.

1.2 Who experiences this

WHO	WHAT IT COSTS THEM TODAY
Brand and agency teams	Manual pre-publication review that does not scale, and no defensible record of having checked. When a claim arrives six months later, there is nothing in the file except a Slack thread.
Marketplaces and platforms	Takedown queues staffed by humans reviewing images with no structured evidence, no consistency between reviewers, and no audit trail for appeals.

WHO	WHAT IT COSTS THEM TODAY
Independent artists and illustrators	No practical mechanism to register a work, prove it existed at a point in time, and have that registration actually consulted by the platforms where reuse happens.
Publishers and newsrooms	Disclosure obligations around synthetic media with no reliable way to determine origin, and reputational exposure when they get it wrong in either direction.

1.3 Why now

Three curves crossed in the last twenty-four months. Generative image models became good enough that origin is genuinely ambiguous to the human eye. Content-provenance standards (C2PA / Content Credentials) matured from a proposal into something shipped by cameras and editing software. And transparency obligations around synthetic media moved from "best practice" into regulation and platform policy.

That combination creates a demand that did not exist before: organisations now need a **structured, auditable, machine-produced record** of what they checked before publishing — not a verdict, a record. Verdicts belong to lawyers. Records belong to infrastructure. Nobody is selling the record.

> THE PROBLEM, IN ONE SENTENCE
>
> Publishers have to decide whether an image is safe to release, the question is really three separate questions with three different kinds of evidence, and the available tools answer them with a single number that is confident, unverifiable and frequently wrong.

02 Solution description

CreatorProof is a pre-publication creative-rights evidence API. It runs three independent AI lanes over a candidate image, keeps their findings separate, applies the customer's own policy to produce PASS, REVIEW or BLOCK, and seals the entire result into a signed evidence packet whose hash is anchored on a public blockchain.

2.1 The pipeline, upload to receipt

01 · INTAKE

Submit

Candidate image posted to `/v1/scans` with a catalog, an intended use and an idempotency key. Returns immediately with a scan ID; detection runs asynchronously.

02 · EVIDENCE

Three lanes

Copy, origin and profile run independently. Each produces typed findings, its own confidence, and an explicit statement of what it was able to look at.

03 · POLICY

Your rules

A versioned policy converts findings into PASS REVIEW BLOCK with recorded reason codes and policy inputs.

04 · PROOF

Sealed receipt

Canonical evidence packet, Ed25519 signature, append-only Merkle log, and a 32-byte commitment attested on chain via EAS.

2.2 The three lanes, and why they stay separate

Lane separation is the core design decision of the product, and it is enforced in code rather than in documentation. The lanes have different data sources, different failure modes and different legal meanings, so they are computed independently, presented independently, and are never averaged together into an "infringement percentage".

LANE 1 — STORED-WORK COPY

A learned copy-detection descriptor nominates candidates from the customer's registered catalog. Classical geometry then attempts to physically align the candidate onto the reference. Aligned pixels are compared. A match requires agreement from **independent** kinds of evidence, not a high similarity score.

Answers: which registered work is present, and where in the frame.

LANE 2 — AI ORIGIN

Cryptographic provenance (C2PA) is checked first, because a signed manifest is worth more than any detector. Then a detector ensemble runs the image under ten stress views. Visible watermark text is read by OCR. Findings are fused into one score that is explicitly labelled **not a probability**.

Answers: what evidence of generative involvement exists — and how strong it is.

LANE 3 — CREATOR PROFILE

A style-specific embedding compares the candidate against enrolled creator profiles, ranked with a density correction so that a "generically popular" profile cannot win by default. A content control checks whether resemblance is really just the same subject matter.

Answers: does this resemble a creator's body of work — advisory only, never authorship.

2.3 Coverage before verdict

Before any lane reports a finding, the scan reports **what it was able to search.** Eligible references, nominated references, verified references, omitted references, failures, the catalog version and a snapshot digest are all recorded. Only a scan whose coverage is `COMPLETE` is permitted to return `NO_MATCH_IN_CHECKED_SOURCES`. Every incomplete scan is visibly review-only.

This is a small rule with a large consequence: **the system is structurally incapable of telling you an image is clean when it did not finish looking.** That single property is what makes the output usable as evidence rather than as reassurance.

2.4 The proof layer

An evidence packet is only useful if a third party can check it without trusting the party who produced it. CreatorProof provides four layers of verifiability, each independently checkable:

LAYER	MECHANISM	WHAT IT PROVES
Canonicalisation	Deterministic JSON serialisation (sorted keys, ASCII, no whitespace) → SHA-256. The proof object is excluded from its own hash.	Two parties hashing the same packet get the same 32 bytes.
Signature	Ed25519 over an RFC 8785 JCS canonicalisation, wrapped in a COSE Sign1 envelope.	The packet came from this issuer and has not been edited.
Transparency log	Append-only RFC 6962-style Merkle tree with signed tree heads and per-leaf inclusion proofs.	This packet was in the log at this tree size, and the log has not been rewritten.
Public anchor	Ethereum Attestation Service `attest()` on Base Sepolia, committing a 32-byte hash.	The commitment existed no later than this block, witnessed by a chain nobody in the transaction controls.

Multi-party co-attestation

The feature we are proudest of architecturally. A verdict from a single vendor is still a single vendor's word. Co-attestation lets a **second, independent organisation** — the agency, the licensor, the marketplace's rights partner — bind its own EVM key to the same evidence commitment.

The counterparty is served a canonical challenge body, signs it with EIP-712 typed data, and the resulting attestation is written on chain with its `refUID` pointing at the original platform attestation. Five gates are enforced before it is accepted: recovered signer address must match the declared signer, the signer must be an active member of the on-chain registry, the recomputed canonical digest must match, the on-chain data must match the body hash, and the reference UID must point at the correct parent.

The result is a decision that is **jointly witnessed**. Neither party can later deny it, and neither party had to trust the other's database.

2.5 What CreatorProof explicitly is not

The boundaries are stated in the product, in the API responses, and in the interface — not buried in terms of service. This is a deliberate product decision, and we consider it a feature rather than a disclaimer.

NOT A LEGAL FINDING

It reports evidence of reuse. Whether that reuse is infringing is a legal question involving licences, fair dealing and jurisdiction that no image model can answer.

NOT RIGHTS CLEARANCE

A PASS means "our checks found nothing against your policy in the sources we searched". It is not permission to publish.

NOT PROOF OF HUMAN ORIGIN

An absent C2PA manifest and a low detector score are not evidence that a person made the work. Absence of evidence is reported as absence of evidence.

2.6 The two portals

ARTIST PORTAL

A creator registers a reference work, states an authorship claim, and receives a registration receipt. Registration runs the AI-origin lane as a gate: works scoring above the configured threshold are refused, with the score and threshold returned in the refusal so the decision is contestable rather than opaque.

This is what makes the catalog worth searching — a registry of works that a human claims and that were screened at intake.

CHECKER PORTAL

Someone about to publish drops in a candidate. They get the bottom line first, then the three lanes, then an evidence workspace that renders the actual geometric correspondences on the pixels — the matched keypoints, the aligned region, the structural agreement.

The workspace exists so a reviewer can disagree with the machine on the basis of the same evidence the machine used.

03 Target users

CreatorProof is API-first and business-to-business. The buyer is an organisation with publication risk; the beneficiary is the creator whose work is in the catalog.

3.1 Primary customers

SEGMENT	WHO SIGNS	THE JOB THEY ARE HIRING US FOR
Creative agencies	Head of production, general counsel	Screen every deliverable before it reaches the client, and hand the client a proof packet with the invoice. Turns a liability conversation into a value-added deliverable.
Brand in-house studios	Brand ops lead, legal ops	Enforce a consistent pre-publication standard across dozens of agencies and freelancers, with an audit trail that survives an internal review.
Marketplaces & UGC platforms	Head of trust & safety	Triage uploads at scale so that human reviewers only see the ambiguous cases, and every takedown or reinstatement has structured evidence behind it.
Publishers & newsrooms	Standards editor, picture desk	Meet synthetic-media disclosure obligations with a defensible record instead of an editor's judgement call.
Licensing & stock platforms	Rights operations	Gate contributor intake, and co-attest licence decisions with the counterparty so both sides hold the same signed record.

3.2 The beachhead

MID-SIZE CREATIVE AGENCIES, 20-200 PEOPLE, RUNNING PAID CAMPAIGNS

We deliberately do not start with everyone. The agency is the right first customer because it has all three properties a wedge needs: **acute pain** (a single infringement claim on a campaign can exceed a year of software spend), **a small buying committee** (a head of production can sign), and **a natural expansion path** (the agency's proof packet is seen by every client it serves, which turns each customer into a distribution channel into a brand).

3.3 THE CREATOR SIDE

Creators are not the paying customer, and pretending otherwise would be dishonest — independent artists are the least able to pay and the most in need. They are the **supply side of the network**. Registration is free to the creator, funded by the checking side.

This is the structural insight that made content-ID systems work: the party who benefits from a match being found is not the party who registered the work. The creator's incentive is that their registry entry is actually consulted by the organisations most likely to reuse their work, and that they hold a timestamped, independently verifiable record of having registered it.

3.4 WHO THIS IS NOT FOR

Consumers looking for a free "is this AI?" button. Anyone wanting a legal opinion. Anyone who needs a crawl of the open internet.

CreatorProof searches declared catalogs, and says so on every result. Being explicit about the boundary is precisely what keeps the output trustworthy inside it — a tool that claims to have checked everywhere is a tool whose "no match" means nothing.

04 Technologies used

4.1 Architecture

A FastAPI backend holds all detection, policy and proof logic behind a provider-boundary abstraction, so every model is a swappable implementation of an interface rather than a call embedded in business logic. A Next.js front end never touches an API key: the browser talks to Next.js route handlers, which talk to the API server-side.

LAYER	TECHNOLOGY
API	Python 3.12 · FastAPI · Pydantic v2 · SQLAlchemy · Alembic · uv (locked dependency graph)
Inference	PyTorch 2.13 · TorchScript · timm · safetensors · OpenCV · Pillow · NumPy
Data	PostgreSQL with row-level security for tenant isolation; SQLite for local development
Web	Next.js (App Router) · React · TypeScript · hand-authored CSS on a single token system · self-hosted variable fonts
Chain	Ethereum Attestation Service · Base Sepolia (chain 84532) · web3.py · eth-account · Solidity · local Ganache harness

LAYER	TECHNOLOGY
Crypto	Ed25519 · COSE Sign1 · RFC 8785 JCS canonicalisation · RFC 6962-style Merkle tree · EIP-712 typed signatures · SHA-256
Provenance	C2PA via `c2patool` 0.27.7 (version and binary digest pinned) · Tesseract OCR 5.3.4
Operations	Prometheus metrics · structured stage timelines · signed webhooks · idempotency keys · append-only usage meters
Clients	First-party Python and TypeScript SDKs, dependency-free

4.2 Production-shaped properties

These are not features a demo needs. They are in the codebase because the intent was always a product, and they are the reason the prototype could be pointed at a real customer without a rewrite:

MULTI-TENANCY

API key resolves to a tenant and a scoped role. PostgreSQL row-level security policies deny access to any query that has not bound a tenant. Break-glass contexts exist for workers and are explicit.

IDEMPOTENCY & ASYNC

Scan creation returns immediately with a recoverable ID. Reusing an idempotency key with changed bytes, catalog or intended use returns a typed `409 IDEMPOTENCY_PAYLOAD_MISMATCH` rather than silently returning a stale result.

METERING

Append-only meters for scans, protected assets, storage bytes, GPU stage seconds, proof anchors and retention tier. Metering failures are logged and swallowed — billing never breaks a customer request.

CRASH-SAFE CHAIN WRITES

Transactions are signed and persisted with their hash *before* broadcast, so a process death mid-send is reconciled rather than duplicated or lost. A dispatcher thread leases jobs and retries.

4.3 API surface

GROUP	ENDPOINTS
Catalog	POST/GET `/v1/works` · POST `/v1/works/bulk` · GET work media · DELETE work
Scans	POST `/v1/scans` · GET scan · GET stages · POST cancel · GET statement · GET statement/verify · GET verification-package
Rights	parties · claims · claim events · licences · licence events · work rights position
Policy	POST/GET `/v1/policies` · POST `/v1/policies/dry-run`
Review	GET `/v1/review-cases` · GET case · POST reviewer actions
Proof	status · integrity · preflight · attestation lookup · transparency checkpoint · consistency · trust-bundle
Network	status · member registry · co-attestation challenge · co-attestations · withdraw
Platform	credentials · webhook endpoints · webhook deliveries · usage

05 AI tools and models

This is the section that matters most, so it is the longest. CreatorProof is not a product with AI added to it. Every question the product answers is answered by a model, and the entire architecture exists to constrain what those models are allowed to conclude.

> **THE DESIGN PRINCIPLE: AI NOMINATES, DETERMINISTIC METHODS VERIFY**
>
> A neural network is superb at "these two images are probably related" and unreliable at "and therefore this is a copy". So in CreatorProof a learned model is never permitted to produce a match on its own. The model **nominates**; classical geometry **verifies** by physically aligning one image onto the other; aligned pixels **corroborate**. A finding requires agreement from independent kinds of evidence. This is why the system can report zero false matches on our negative set while still recovering mirrored, cropped and re-compressed reposts.

5.1 Model inventory

MODEL	ARCHITECTURE	RUNS	ROLE
SSCD `disc_mixup`	ResNet-50 family, 512-d descriptor	Local · 95 MB	Copy-detection retrieval. Nominates catalog works that may be present in the candidate.
CSD ViT-L	Vision Transformer Large, 768-d style descriptor	Local · 2.3 GB	Creator-profile resemblance. Encodes style independently of subject matter.
Community Forensics	ViT-S/16 at 384px, binary head	Local · 84 MB	Generalising synthetic-image detector. Trained across a very large set of generator families.
Sightengine `genai`	Vendor model	Cloud · optional	Second opinion on AI origin when credentials are configured. The system runs fully without it.
Tesseract OCR	LSTM text recognition	Local · CLI	Reads visible "AI generated" watermark text that generators burn into the frame.
`c2patool`	Standards implementation	Local · CLI	Validates cryptographic content credentials and reads signed AI-generation assertions.

All three neural models are present on disk, verified against pinned SHA-256 digests in the model bundle manifest, and confirmed to load and run. None of them is a placeholder.

5.2 Lane 1 — Copy detection

SSCD (SELF-SUPERVISED COPY DETECTION)

From Meta AI Research, trained specifically for the copy-detection task rather than for general similarity. This distinction matters: a CLIP-style embedding will happily report that two different photographs of a sunset are similar. SSCD is trained to be invariant to the transformations that reposts actually apply — re-compression, rescaling, cropping, colour shifts, overlays — while remaining discriminative between genuinely different works.

Preprocessing	RGB, shortest side resized to 288px bilinear, ImageNet normalisation
Output	512-dimensional L2-normalised descriptor; similarity is a cosine dot product
Regional retrieval	Whole image plus up to five overlapping crops at 0.64 area fraction, so a work occupying a corner of a collage is still nominated. Non-whole views carry a small score penalty.
Caching	Content-addressed embedding cache keyed by provider identity digest and source SHA-256

GEOMETRIC VERIFICATION — THE PART THAT MAKES IT EVIDENCE

Nomination is followed by SIFT keypoint detection (ORB as fallback), mutual nearest-neighbour matching under a ratio test, and homography estimation with USAC_MAGSAC. If a homography is found, the candidate can be *physically aligned* onto the reference, and the system knows exactly which pixels correspond to which. Those correspondences are what the evidence workspace draws on screen.

Two escalation paths were added after benchmarking exposed specific, reproducible failures:

- **Mirror recovery.** SIFT descriptors are not reflection-invariant, so a horizontally flipped repost — one of the most common evasions — failed completely: 0 of 14 mirrored cases were detected, despite the descriptor scoring them as near-identical. The verifier now retries with a mirrored query when descriptor similarity is high, and composes the reflection into the homography so the alignment still maps back onto the file as it was submitted. Result: **14 of 14 recovered.**
- **Corroboration-required alignment.** Flat, repetitive artwork — logos, vector illustration, graphic design — defeats the ratio test, because a repeated visual element has many equally good matches and every one of them is discarded. A relaxed matching pass now runs for these cases, but its output is graded `CORROBORATION_REQUIRED` and cannot produce a match unless aligned-pixel structural agreement independently confirms it.

ALIGNED PERCEPTUAL AGREEMENT

Once aligned, the two images are compared where they overlap: luminance correlation, gradient correlation, gradient-magnitude similarity, and SSIM, combined into a structure-consensus figure. This is the pixel-level second opinion that makes the relaxed geometry path safe.

FUSION

A rule-based fusion layer takes descriptor similarity, perceptual hash distance, geometry grade and aligned structure and decides between `MATCH_FOUND`, `REVIEW` and `NO_MATCH_IN_CHECKED_SOURCES`, emitting typed reason codes throughout. It is deliberately rule-based rather than a learned classifier, because the fusion step is the one a customer will be asked to justify, and "the model said so" is not a justification.

5.3 Lane 2 — AI origin

The hardest lane, and the one where honesty is a technical requirement rather than a disclaimer. There is no universal AI-image detector. Anyone who claims otherwise is selling something. The architecture is built around that constraint.

EVIDENCE HIERARCHY

1. **Cryptographic provenance (C2PA).** If the file carries a validly signed manifest asserting `trainedAlgorithmicMedia` from a trusted signer, that is the answer and no detector is needed. Signature validation, signer trust and the ingredient chain are all evaluated. Crucially, an *absent* manifest is reported as `NOT_PRESENT` — unknown, not human.
2. **Detector ensemble.** Community Forensics locally, Sightengine in the cloud if configured. The two are never blended: a healthy cloud result is primary, and local detectors are the fallback, so a result is always attributable to a known source.
3. **Visible markers.** Tesseract OCR across five crops reads burnt-in generator watermarks. Treated as a forgeable review signal, never as provenance.
4. **Spectral diagnostics.** FFT frequency ratios, Laplacian variance, residual channel correlation. Descriptive context for a human reviewer; never permitted to decide origin.

MULTI-VIEW STRESS TESTING

A detector that changes its mind when an image is saved as JPEG is not measuring the image, it is measuring the compression. Every local detector therefore evaluates ten views: the original, four delivery transforms (JPEG q95, JPEG q75, 72% downscale-restore, light Gaussian blur), and five spatial crops. The spread across those views becomes a **transform-stability** figure that down-weights unstable detectors in the fusion, and the original's score is never diluted by its own stress views.

THE FUSED DETECTOR SCORE

Detector outputs are grouped by *evidence family* — the kind of artefact they exploit — and aggregated in logit space with reliability weights derived from calibration status and transform stability. Families are then fused into a single `fused_detector_score`. Its schema name is deliberately unlovely:

```
// the label attached to every fused score the system emits
ORIGINAL_PRESERVING_QUALITY_WEIGHTED_ENSEMBLE_NOT_PROBABILITY
```

It is not a probability that the image is AI-generated, and the system will not let a caller pretend it is. The highest-authority outcome, `LIKELY_AI_GENERATED`, additionally requires two independently verified *and calibrated* evidence families to agree, with stability across views. Today only one family is registered as verified, so the system **abstains from its own strongest claim** — by design, and visibly, rather than quietly rounding up.

REGISTRATION GATE

In the Artist portal this score becomes an admission control. A work whose fused score exceeds the configured threshold (default 0.50) is refused registration, and the refusal returns the score and the threshold so the artist can see and contest the decision. Inconclusive findings never block. The catalog is a registry of human-claimed work, and the gate is how it stays one.

5.4 Lane 3 — Creator profile

CSD (Contrastive Style Descriptors, University of Maryland) produces a 768-dimensional embedding trained to capture artistic style while discarding subject matter — so two paintings of completely different scenes by the same artist land near each other, which a general embedding will not do.

- **Profiles** are built by averaging the CSD embeddings of works grouped by claimant, forming a creator anchor.
- **Ranking uses CSLS**, a local-density correction borrowed from cross-lingual word retrieval, which stops a profile that sits in a dense region of style space from winning every comparison by default — the hubness problem.
- **A content control** runs SSCD against the same works, so the system can distinguish "same style" from "same subject rendered similarly".
- **Support is catalog-relative** and empirical, computed with Bonferroni-adjusted tail probabilities against the actual catalog, and labelled as such rather than presented as a universal confidence.
- **The lane is advisory.** It is structurally prohibited from escalating a policy outcome on its own, because resemblance is not evidence of copying.

5.5 AI governance — the unusual part

Most projects ship models. This one ships a mechanism for being accountable about models, which we think is the more interesting contribution:

MECHANISM	WHAT IT ACTUALLY ENFORCES
Model bundle manifest	Every model artefact is pinned by SHA-256, alongside the Python version, the locked dependency digest, and a hash of the application source tree. Validated at container construction. A silently swapped weight file is a startup-time failure, not a mystery in production.
Qualification ladder	Bundles are graded `SOURCE_VERIFIED` → `RUNTIME_READY` → `SMOKE_TEST_ONLY` → `DEMO_READY` → `CALIBRATED_DOMAIN_READY` → `PRODUCTION_READY`. Ours self-reports as **RUNTIME_READY**. We did not promote it to flatter ourselves.
Evidence-family registry	A detector only counts toward independence requirements if its model version, artefact digest and preprocessing match a registered family. An unregistered detector still runs, but cannot be used to satisfy a "two independent families agree" gate.
Calibration registry	Platt-scaling parameters per domain, requiring minimum sample counts and matching digests. Currently absent, so every lane is marked `NOT_CALIBRATED_FOR_TARGET_DOMAIN` and high-authority outcomes are suppressed.
Acceptance policy	Benchmark gates a model must clear for promotion, with `automatic_promotion_allowed: false` — a human promotion record is required. The policy's own ratification state is `DRAFT_NOT_RATIFIED`, and it says so.
Drift monitoring	Compares live score distributions against recorded baselines and raises revalidation warnings on standardised shift.

WHY THIS IS THE ANSWER TO "IS AI USED PURPOSEFULLY?"

Purposeful use of AI is not "we called a model". It is knowing precisely what each model can and cannot support, encoding that knowledge as a constraint the code enforces, and being willing to abstain when the evidence does not reach the bar. Every one of the mechanisms above exists to make a model *less* able to overclaim. That is the opposite of bolting AI on.

5.6 AI tools used to build CreatorProof

Disclosed in full, because how a small team produced a system this size is part of the honest answer:

TOOL	USED FOR
Cursor (agentic coding)	Primary development environment. Implementation across the API, inference pipeline, chain integration and interface; large-scale refactors; test authoring.
Frontier LLMs via Cursor	Architecture review, adversarial critique of our own claims, threat modelling on the proof layer, and drafting of the honest-scope language that constrains the product.
AI-assisted benchmarking	Building the copy-detection benchmark harness, diagnosing the mirror and flat-artwork failures from its output, and validating the fixes.
ElevenLabs	Narration for the video pitch.
Generative video	Short conceptual b-roll clips in the pitch video, clearly used as illustration between real screen recordings.

Every architectural decision, threshold, scope boundary and honesty constraint described in this document was a human decision. AI accelerated the writing of the code; it did not choose what the product refuses to claim.

06 Evidence that it works

6.1 Copy-detection benchmark

We built a harness that takes 14 source images, applies 26 transformations that model how images are actually reused in the wild, and runs each result through the complete production pipeline — retrieval, geometry, aligned perceptual, fusion. Negatives are unrelated works and hard crops against different works.

364	98.08%	99.45%	0
positive cases across 26 transformations	returned `MATCH_FOUND`	flagged as match or review	false matches on 115 negatives

TRANSFORMATION FAMILY	CASES	MATCH	NOTE
Re-encoding — JPEG q80 / q55 / q35, identical re-encode	56	56	Perfect
Rescaling — 30% / 50% / 150%	42	42	Perfect

TRANSFORMATION FAMILY	CASES	MATCH	NOTE
Colour — brightness, contrast, saturation, greyscale	56	56	Perfect
Repost formats — caption bar, letterbox, screenshot, social repost	56	56	Perfect
Mirroring — horizontal flip	14	14	Was 0/14 before the mirror-recovery fix
Rotation — 90°, 10°, 2°	42	40	Two flat-artwork cases fall to review or miss
Cropping — 90% / 75% / 60% / 45%	56	53	Aggressive crops of repetitive artwork are the residual weakness
Compound — crop+JPEG, scale+rotate, blur	42	40	—

WHAT WE STILL FAIL, STATED PLAINLY

Two of 364 cases miss outright, and both are the same pathological source: a flat, highly repetitive graphic with almost no distinctive keypoints, cropped to 60% or rotated 10°. SIFT cannot find enough distinct features on it. We know exactly why it fails and could raise recall by loosening thresholds — and chose not to, because the same loosening introduces false-match risk, and a false match in this product is far more damaging than a referral to human review. Five further cases return *review* rather than *match*, which is the system behaving as intended: uncertain, and saying so.

6.2 Test suite

400	47	9.5k
automated tests, all passing	test suites	lines of test code

Coverage includes adversarial cases for the offline verifier, tamper and replay attacks on co-attestation, crash-recovery for chain writes, Merkle inclusion and consistency proofs, Ed25519 and COSE test vectors, idempotency conflict semantics, tenant isolation, and the detection-fusion behaviours described above.

6.3 Blockchain status — stated precisely

CAPABILITY	STATE	DETAIL
EAS attestation pipeline	IMPLEMENTED	Preflight, gas estimation, signing, crash-safe persist-before-broadcast, receipt parsing, confirmation depth, on-chain read-back verification.
Three attestation schemas	IMPLEMENTED	Evidence packet, transparency checkpoint, counterparty attestation — each with its own revocability policy.
EIP-712 co-attestation	IMPLEMENTED	Real secp256k1 signature verification with five acceptance gates; tested against tampering, replay and staleness.
Merkle transparency log	IMPLEMENTED	Append-only, signed tree heads, inclusion and consistency proofs. Labelled *not blockchain* everywhere it appears.

CAPABILITY	STATE	DETAIL
Offline verification	IMPLEMENTED	Standalone script verifies a statement with no network access, against an externally pinned issuer fingerprint.
Live Base Sepolia transactions	REQUIRES SETUP	Needs a funded key, registered schema UIDs and RPC configuration. Default development mode anchors to the local Merkle log and **says so in the interface** rather than implying a chain write.
Clearance-receipt token	CONTRACT ONLY	Solidity exists and deploys locally; no API pipeline mints it. Not claimed as a product feature.

07 Judging criteria alignment

Our own assessment against the published criteria, including where we are weak. We would rather a judge hear our weaknesses from us than find them.

Problem fit and real-world impact 25%

The problem is defined more precisely than the market defines it — we separate three questions that competitors merge, and we can articulate the specific harm each conflation causes. It is acutely felt by identifiable buyers with budget. Measurable value is credible (review hours displaced, claims avoided, audit trail produced) but **not yet measured**: we have no pilot data or letters of intent. That is the honest gap.

Innovation and idea quality 20%

The innovation is architectural rather than a single novel algorithm: separated evidence lanes, coverage-before-verdict, AI-nominates-geometry-verifies, and multi-party co-attestation as a trust primitive. AI use is unusually purposeful — the governance layer exists specifically to stop models overclaiming, and the system visibly abstains from its own strongest classification because the calibration evidence is not there.

Technical implementation 20%

A working system: 400 passing tests, three neural models running locally, a measured benchmark with documented failures, a complete chain integration with crash-safe writes. Weaknesses we accept: the benchmark is ours rather than a public dataset, detector calibration is absent, and a live testnet transaction requires operator configuration rather than being pre-baked in the repository.

Product potential and scalability 20%

Unusually productised for a hackathon build — multi-tenancy with row-level security, metering, quotas, webhooks, idempotency, first-party SDKs. A named beachhead and a network-effect flywheel on the catalog. Weakness: no customer discovery interviews and no validated pricing. The business model is reasoned rather than tested.

Demo and presentation 15%

The demo shows the real product doing real work: register a work, scan a mirrored crop of it, watch geometry recover the alignment, see the three lanes stay separate, open the receipt. The risk we are managing is that our honesty — "this is not a legal finding" — can be misheard as weakness. The pitch is written to make restraint read as credibility, which is what it is.

08 Limitations and honest scope

Reproduced from the product's own documentation, because these constraints are enforced in the code and shown in the interface, not tucked into a footer.

WHAT IT DOES

- Searches a declared catalog and reports what matched
- Reports evidence of generative involvement
- Reports style resemblance as advisory context
- Applies the customer's policy and records why
- Proves a packet existed, unaltered, at a point in time
- Reports coverage honestly, every time

WHAT IT DOES NOT DO

- Determine copyright infringement
- Certify a work as human-made or AI-free
- Establish authorship or ownership
- Provide legal advice or clearance
- Prove the contents of a packet are true
- Search the open internet

Technical limitations we disclose

- **No universal AI detection.** Scores are labelled not-probability, high-authority outcomes require calibrated independent families, and the system abstains when they are unavailable.
- **Absence of a C2PA manifest is not evidence of human origin.** It is reported as unknown.
- **Style resemblance is not attribution.** The lane cannot escalate policy on its own.
- **Thresholds are prototype operating points**, chosen against our own benchmark, not ROC-calibrated against a customer corpus.
- **Catalog truncation is real.** Above the configured reference limit, a scan cannot claim complete no-match and is marked review-only.
- **A local Merkle receipt is not a blockchain anchor**, and is labelled as such. Only a mined, validated EAS receipt is presented as a public-chain anchor.
- **The model bundle is `RUNTIME_READY`, not production-qualified.** Promotion requires customer-corpus benchmarks and a human sign-off we have not manufactured.

WHY WE ARE TELLING YOU THIS

Because a system that produces evidence has to be more careful than one that produces opinions. Every limitation above is a place where a less careful product would have shipped a confident number instead. The restraint *is* the product.